

Robustness to Prior Misregistration: the Trust Gate

Patient-Specific Residual INR — SLAM Ax_T2_2D, change_extent=0

July 8, 2026

Setup

A well-registered follow-up slice is reconstructed from 50% of its pixels (random). The prior is deliberately shifted by s pixels before use, so the *only* source of prior error is misregistration. A method that over-trusts the prior ghosts the shifted anatomy into unobserved regions. **current-only INR** uses no prior (robust floor); **prior-finetune** is NeRP-style; **ours** adds a frozen prior; **ours+gate** adds a spatial trust gate $g(c) \in [0, 1]$ that opens where the prior conflicts with the data.

Table 1: Reconstruction vs. prior shift s (px) at 50% supervision. PSNR/SSIM higher = better; PBS ≈ 1 preserves structure.

shift	method	PSNR	SSIM	NMSE	PBS
0	prior-copy	24.22	0.826	0.0707	0.000
	current-only INR	31.01	0.910	0.0148	1.066
	prior-finetune (NeRP)	32.02	0.927	0.0117	1.051
	prior+residual (ours)	30.99	0.885	0.0149	0.949
	prior+gated residual (ours+gate)	31.36	0.941	0.0137	0.931
4	prior-copy	16.19	0.509	0.4483	2.526
	current-only INR	30.98	0.911	0.0149	1.065
	prior-finetune (NeRP)	31.17	0.921	0.0143	1.067
	prior+residual (ours)	27.72	0.807	0.0316	1.187
	prior+gated residual (ours+gate)	28.55	0.908	0.0261	1.140
8	prior-copy	15.37	0.481	0.5422	2.997
	current-only INR	31.18	0.908	0.0142	1.079
	prior-finetune (NeRP)	30.87	0.917	0.0153	1.074
	prior+residual (ours)	28.33	0.790	0.0274	1.213
	prior+gated residual (ours+gate)	28.69	0.895	0.0252	1.125
12	prior-copy	15.09	0.460	0.5780	3.231
	current-only INR	31.01	0.913	0.0148	1.070
	prior-finetune (NeRP)	30.73	0.913	0.0158	1.078
	prior+residual (ours)	28.15	0.759	0.0286	1.218
	prior+gated residual (ours+gate)	28.77	0.882	0.0248	1.134
16	prior-copy	14.65	0.431	0.6404	3.562
	current-only INR	30.92	0.909	0.0151	1.089
	prior-finetune (NeRP)	30.69	0.908	0.0159	1.086
	prior+residual (ours)	28.57	0.760	0.0259	1.204
	prior+gated residual (ours+gate)	28.87	0.868	0.0242	1.153

Table 2: Mean gate opening $\bar{g}$ vs. prior shift (higher = the gate trusts the prior less). The gate should open as misregistration grows.

shift (px)	0	4	8	12	16
$\bar{g}$	0.032	0.065	0.073	0.078	0.084

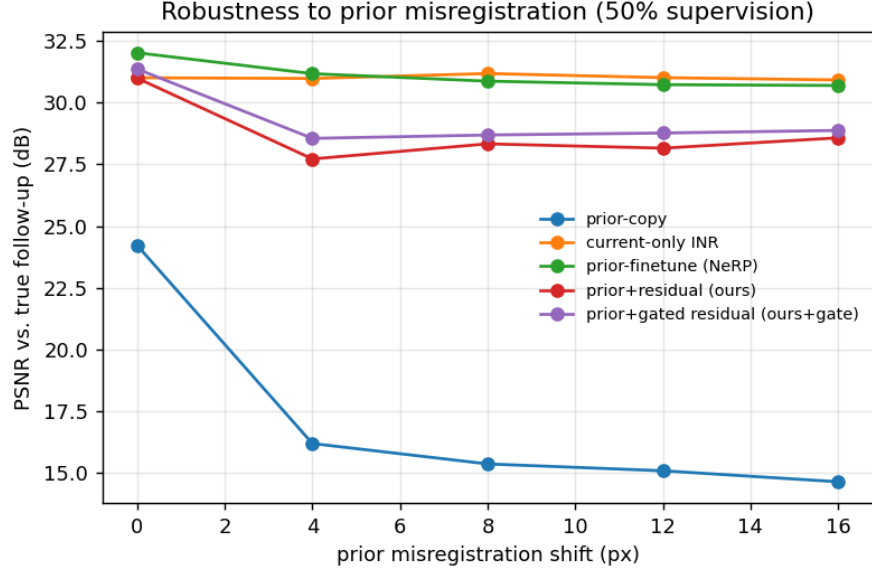


Figure 1: PSNR vs. prior shift. current-only is flat (no prior); naive prior methods degrade as the prior worsens; the gate degrades more gracefully.

Qualitative panels

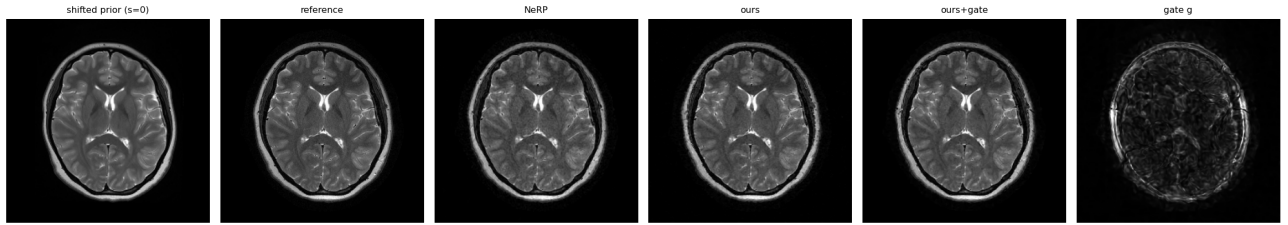


Figure 2: Prior shift $s = 0$ px. Left to right: shifted prior, reference, NeRP, ours (frozen+residual), ours+gate, and the gate map g (white = open = distrust prior).

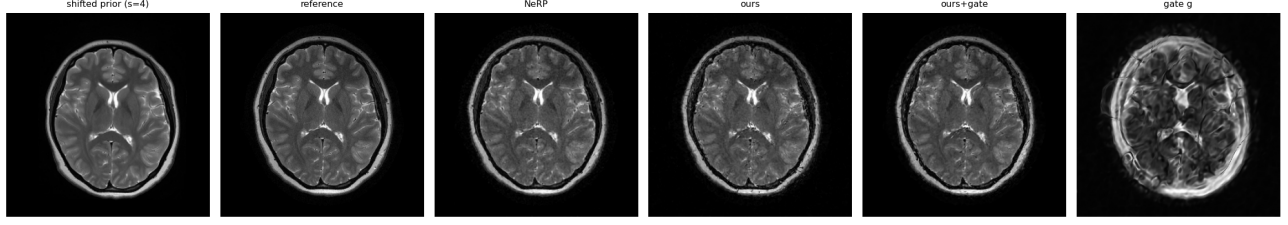


Figure 3: Prior shift $s = 4$ px. Left to right: shifted prior, reference, NeRP, ours (frozen+residual), ours+gate, and the gate map g (white = open = distrust prior).

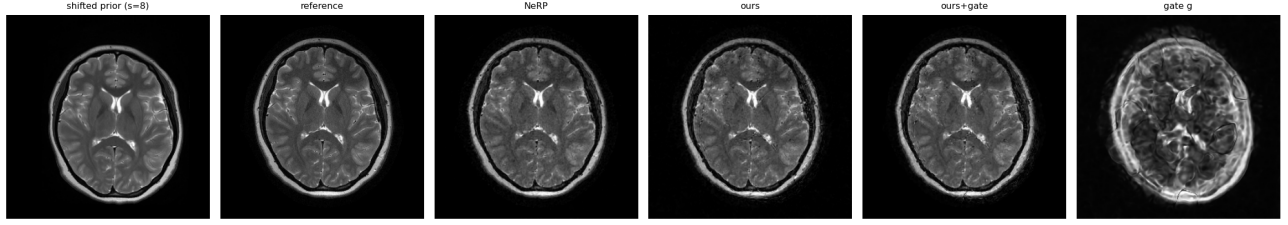


Figure 4: Prior shift $s = 8$ px. Left to right: shifted prior, reference, NeRP, ours (frozen+residual), ours+gate, and the gate map g (white = open = distrust prior).

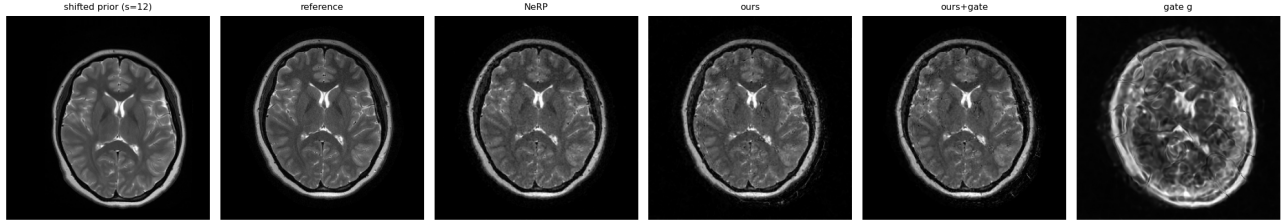


Figure 5: Prior shift $s = 12$ px. Left to right: shifted prior, reference, NeRP, ours (frozen+residual), ours+gate, and the gate map g (white = open = distrust prior).

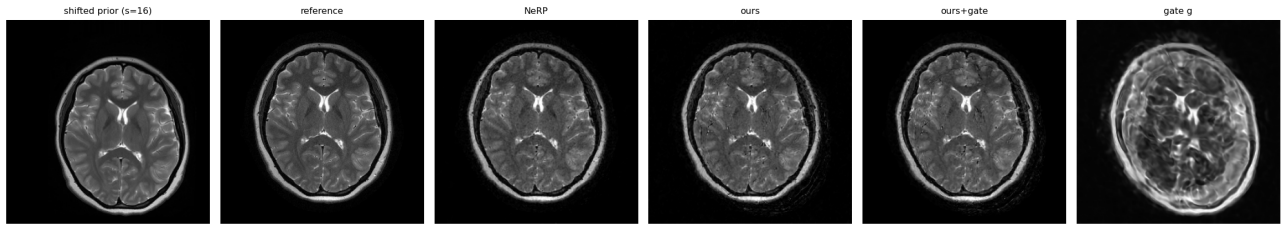


Figure 6: Prior shift $s = 16$ px. Left to right: shifted prior, reference, NeRP, ours (frozen+residual), ours+gate, and the gate map g (white = open = distrust prior).